

(KTXFI2EBNF) Physics II. Exam 3
January 19, 2026

I. Moving Reference Frames. Special Theory of Relativity

1. Write down the Lorentz transformation equations used in the special theory of relativity. Specify the exact physical meaning of the symbols used in the formulas, including the reference frame in which they are defined. (5 points)
2. State the postulates formulated by Albert Einstein for the special theory of relativity. (Pay special attention to the precise and exact formulation of the postulates.) (5 points)

II. Limits of the Classical Conceptual Framework

3. Formulate the de Broglie principle in your own words. (5 points)
4. What is the Compton effect, what did it prove, and how was this demonstrated? (5 points)
5. What is meant by the dual nature of electromagnetic radiation? Describe the meaning of this expression. (5 points)

III. Elements of Quantum Mechanics. Physics of Condensed Matter. Theory of Electrical Conduction

6. What is a forbidden band (band gap)? Define the concept. (5 points)
7. Write down the one-dimensional, time-independent Schrödinger equation using the standard notation. Specify the exact physical meaning of the symbols appearing in the equation. (5 points)

IV. Development of Atomic Theory

8. State the Pauli exclusion principle. (5 points)
9. The meaning of the spin quantum number. What does the spin quantum number represent? Describe in detail what is known about its physical meaning. What values can it take? (5 points)
10. Given the element with atomic number 10 in the periodic table, write down its electronic configuration using the standard atomic physics notation. (5 points)

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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